

How to Learn Effectively: An Evidence-Based Guide for Masters Students

TL;DR

- The techniques with the strongest experimental support are **retrieval practice** (self-testing; testing effect typically $d/g \approx 0.50-0.70$) and **spaced/distributed practice** — Dunlosky et al. (2013) rate both "high utility." These should be the backbone of your routine, while the methods most students actually rely on — re-reading, highlighting, and cramming — are rated **low utility**.
- **Interleaving, elaboration/self-explanation ($g \approx 0.55$), dual coding, and worked examples** have moderate-to-good support but with important conditions (interleaving works best for related, confusable categories; worked examples lose their value as you gain expertise — the "expertise reversal effect").
- **Learning styles** (matching visual/auditory/kinaesthetic instruction to a learner's "style") and the **learning pyramid / fixed retention percentages** are myths with no credible evidence. The deeper organising insight is Bjork's distinction between **learning and performance**: what feels effective (fluency) is a poor guide to durable learning.

Key Findings

The single most important reframing for a postgraduate learner is **Bjork's learning-performance distinction**: *performance* is how well you can do something during study today; *learning* is the relatively permanent change measured later, in a new context. Most popular study habits (re-reading, highlighting, cramming) boost short-term performance and the *feeling* of fluency while doing little for long-term retention. The techniques that work — retrieval, spacing, interleaving — feel harder and slower, which is why Bjork calls them "desirable difficulties." Crucially, your subjective sense of how well you're learning (your "judgment of learning") is systematically biased by fluency and skews toward overconfidence, so you cannot trust the feeling of ease as a signal of learning.

Two techniques have genuinely strong, replicated evidence: **retrieval practice** and **spaced practice**. The others fall on a spectrum from "moderate and conditional" (interleaving, elaboration, self-explanation, dual coding, worked examples) to "popular but weak" (re-reading, highlighting, summarising) to "actively debunked" (learning styles, the learning pyramid).

Details

1. Techniques with STRONG evidence

Retrieval practice / the testing effect. Actively recalling information from memory (self-quizzing, flashcards, closed-book brain-dumps) produces more durable learning than

restudying. This is the best-supported technique in the field. In Roediger & Karpicke's landmark 2006 study (*Psychological Science* 17(3)), students who studied a passage once then took a test recalled **56% after a one-week delay, versus 42%** for students who studied twice ($d = 0.83$). In their second experiment, repeated testing beat repeated studying **61% vs 40%** on a one-week delayed test — even though the testing group read the passage only 3.4 times versus 14.2 times for the study group. Tellingly, on an immediate (5-minute) test the ordering reversed (repeated study 83% vs repeated testing 71%), and the repeated-study group was the *most confident* they would remember — a vivid demonstration of the performance/learning gap. (wustl)

Meta-analyses confirm robustness. Rowland (2014) found Hedges' $g = 0.50$ comparing testing to restudy. Adesope, Trevisan & Sundararajan (2017, *Review of Educational Research*) analysed 272 independent effects from 188 experiments and found practice testing produced a moderate, significant weighted mean effect of **+0.51 vs restudying** (and +0.93 vs no activity); effects were slightly larger in classroom settings ($g = 0.67$) than labs ($g = 0.62$), and single practice tests ($g = 0.70$) outperformed multiple tests ($g = 0.51$), with multiple-choice practice (+0.70) beating short-answer (+0.48). Agarwal, Nunes & Blunt (2021, *Educational Psychology Review*) reviewed applied classroom research specifically: across **50 experiments (from 37 studies, $n = 5,374$)**, the **majority (57%) showed medium or large benefits**, concluding educators should implement retrieval practice "with less concern about the precise format or timing." Karpicke & Blunt (2011, *Science*) showed retrieval practice even beat concept mapping (an elaborative technique) on later tests of comprehension and inference — and students wrongly predicted concept mapping would be better. (TeacherToolkit + 2)

How to apply: After reading a chapter or attending a lecture, close the material and write down everything you can recall ("brain dump"), then check. Use flashcards (e.g. Anki) for terminology and frameworks. Convert lecture notes into questions and answer them from memory. Do past exam questions and practice problems without notes. The effortful struggle to retrieve is the mechanism — don't peek too early.

Spaced (distributed) practice vs massed practice (cramming). Spreading study over time produces far better long-term retention than the same total time crammed into one session. Cepeda et al. (2006, *Psychological Bulletin*) meta-analysed 839 assessments across 317 experiments (184 articles) and found a reliable spacing benefit; their follow-up work suggests the optimal gap between sessions scales with how long you need to retain the material — roughly **10–20% of the retention interval** (e.g. to remember something for a year, review roughly every 1–2 months). Dunlosky et al. (2013) rate distributed practice "high utility." Spacing is most powerful when *combined* with retrieval — "spaced retrieval practice" or successive relearning. (Augmentingcognition)

How to apply: Plan study across the whole term, not the week before exams. Schedule short, repeated review sessions of the same material days or weeks apart. Use a spaced-repetition

flashcard system that automatically schedules reviews. Deliberately revisit "old" topics while studying new ones.

Interleaving vs blocking. Mixing different problem types or topics within a session (ABCABC) rather than practising one to mastery before moving on (AAABBBCCC) improves the ability to discriminate between problem types and choose the right approach. The evidence is good but more conditional than retrieval or spacing. Brunmair & Richter's (2019) meta-analysis (59 studies, 238 effect sizes) found an overall Hedges' $g = 0.42$, with strong moderation by material: interleaving was most beneficial for visually similar, confusable categories — paintings $g \approx 0.67$ — and smaller for mathematics tasks ($g \approx 0.34$), with some studies showing no effect or harm depending on implementation. A systematic review by Firth, Rivers & Boyle similarly found memory benefits of up to $g = 0.65$ and transfer benefits up to $g = 0.66$, greatest "when differences between items are subtle." Classroom durability is less certain: a year-long randomised trial assigning 62 classrooms in Nigeria to blocked vs interleaved maths problem sets (van der Haar, Gray-Lobe, Kremer & de Laat, J-PAL/IGC working paper AI23-876) found interleaving raised short-term retention by **0.29 SD** but found "no evidence that interleaving improves average performance on a cumulative assessment measuring retention of material over the academic year." Part of interleaving's benefit overlaps with spacing (it naturally distributes practice).

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How to apply: Mix problem types when practising (especially in quantitative subjects like statistics, economics, finance). Interleave related-but-confusable concepts (competing theories, similar diagnostic categories, contrasting case types) so you learn to tell them apart. Don't interleave totally unrelated material — the benefit comes from discriminating between things you might otherwise confuse.

Elaboration / elaborative interrogation / self-explanation. Elaboration means asking "why" and "how" and connecting new material to what you already know; self-explanation means explaining to yourself how a problem is solved or why a fact is true. Bisra et al. (2018, *Educational Psychology Review*) meta-analysed self-explanation: 69 effect sizes from 64 studies (~5,917 participants), random-effects mean $g = 0.55$, with benefits across subject areas, conceptual and procedural knowledge, and education levels. Dunlosky et al. (2013) rate both elaborative interrogation and self-explanation "moderate utility." These are particularly suited to postgraduate work because they build the integrated, transferable understanding that Masters assessment demands. One caveat: self-explanation aimed purely at one's own mental state ("metacognitive explanation") is less effective than explaining the material itself.

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How to apply: While reading, repeatedly ask "Why is this true? How does this connect to X? Why does this step follow?" After working a problem, explain each step aloud as if teaching someone. Relate new theories to ones you already know and articulate the differences.

Dual coding. Combining words with relevant visuals (diagrams, graphs, timelines, process maps) can aid learning because verbal and visual information are processed through partially separate channels (Paivio's dual-coding theory; Mayer's Cognitive Theory of Multimedia Learning). The benefit is real but *conditional*: words and images must be conceptually integrated, temporally aligned, and not redundant. Mayer's multimedia principles (e.g. the modality principle — spoken word + image often beats on-screen text + image; the coherence principle — remove decorative clutter) define when it helps and when it overloads. Butcher (2006) reported diagrams added to text improved understanding by ~0.48 SD. Adding decorative or redundant images does not help and can hurt. (ERIC) (Structural Learning)

How to apply: Build your own diagrams, concept maps, and flowcharts that genuinely map relationships — don't just decorate notes. Pair a verbal explanation with a clear visual of the same idea. Avoid duplicating identical text and narration simultaneously.

Worked examples and the expertise reversal effect. For *novices* in a topic, studying fully worked examples (step-by-step solutions) is more efficient than unguided problem-solving, because it reduces the cognitive load of floundering through means-ends search (Sweller). But this advantage *reverses* as expertise grows: Kalyuga, Ayres, Chandler & Sweller (2003) documented the "expertise reversal effect" — techniques that help novices (detailed worked steps, extra explanations) become redundant or even detrimental for more knowledgeable learners, who learn more by solving problems themselves. This is highly relevant to Masters students, who are often expert in some sub-areas and novice in others. (Springer)

How to apply: When entering a genuinely new topic, study worked solutions carefully and self-explain each step before attempting problems. As you gain fluency, "fade" the support — switch to completion problems (partly worked) and then full independent problem-solving. Match the guidance to your current expertise in *that specific* topic.

2. Techniques with WEAK or LOW evidence (that students use anyway)

Re-reading. The most commonly reported strategy and one of the least efficient. Benefits are small, inconsistent, and quickly diminishing after the first re-read; re-reading mainly increases familiarity, which fuels the *illusion of fluency* (you mistake "this looks familiar" for "I know this"). Dunlosky et al. (2013) rate it "low utility." It is far weaker than spending the same time on retrieval. (TILT)

Highlighting / underlining. Rated "low utility" by Dunlosky et al. (2013). On average it offers little benefit beyond plain reading, and because it draws attention to isolated facts it can actually *harm* performance on higher-level tasks requiring inference and connecting ideas. Most students also over-highlight. It can have limited value if done very selectively and actively (e.g. Yue et al. 2015), but as typically used it is close to passive reading. (nih + 3)

Summarising. Rated "low utility" overall, but with a real caveat: summarising *can* work for students already skilled at producing good summaries, and it shares mechanisms with effective elaboration when it forces selection and integration rather than copying. Quality and training matter enormously, so for most students it is unreliable without instruction. (Wordpress)

Massed practice / cramming. Produces decent short-term performance (hence its popularity before exams) but poor long-term retention — exactly the performance/learning trap. Survey evidence (Karpicke et al. 2009; Hartwig & Dunlosky 2012; Blasiman et al. 2017) shows students *intend* to space but default to cramming under deadline pressure, and that massing is associated with using fewer effective strategies overall. Cramming is not useless for an exam tomorrow, but it builds little that lasts — a serious problem for cumulative postgraduate knowledge. (PubMed)

3. Popular myths NOT supported by evidence

Learning styles (visual/auditory/kinaesthetic matching). The belief that matching instruction to a learner's preferred "style" improves learning — the "meshing hypothesis" — is not supported by evidence. Pashler, McDaniel, Rohrer & Bjork (2008) reviewed the literature and found almost no studies used the required experimental design (a crossover interaction), and those that did generally *failed* to find the predicted effect. Subsequent reviews and experiments (e.g. Rogowsky et al. 2015/2020; Cuevas 2015) reach the same conclusion. People do have *preferences*, and material has a "best" modality (you learn geography from maps regardless of "style"), but tailoring teaching to supposed styles does not improve outcomes. Belief in learning styles nonetheless remains very high among educators — a genuine "neuromyth." (PubMed + 2)

The "learning pyramid" / fixed retention percentages. The claim that we retain "10% of what we read, 20% of what we hear ... 90% of what we teach others" is fabricated. The specific percentages trace to no verifiable study; they were misattributed to Edgar Dale's "Cone of Experience," which contained *no numbers* and made no retention claims. Masters (2013, 2020; *Medical Teacher / Medical Education*) and Letrud (2012) document the myth's spread and total lack of empirical basis. Treat any source citing these percentages as unreliable. (Groklopedia)

(Taylor & Francis Online)

Other debunked/over-stated claims. Be sceptical of: pure discovery learning / minimal guidance (inefficient for novices — guidance helps); the idea that "germane load" is a separate, freely addable resource (cognitive load theorists have themselves revised this, folding germane processing into intrinsic load); and the notion that immediate performance gains equal learning (the core fluency trap).

4. Supporting cognitive science for adult learners

Cognitive load theory (Sweller). Working memory is the bottleneck of learning. CLT distinguishes *intrinsic* load (inherent complexity of the material, which interacts with your prior

knowledge), *extraneous* load (load from poor presentation — split attention, clutter, redundancy — which you should minimise), and *germane* load (effort devoted to building schemas; recent formulations fold germane load into intrinsic processing rather than treating it as a separate, addable source). The practical upshot: reduce extraneous load (clean materials, integrated diagrams, defined terms) so capacity is free for the genuinely hard thinking.

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Working memory limits. Working memory holds only a few items at once — Miller's famous "7 ± 2" was a rough estimate that later work revised downward; Cowan (2001) argues the true capacity, when rehearsal is controlled, is closer to **four chunks**. Capacity is effectively expanded by *chunking* — grouping items into meaningful units — which depends on prior knowledge stored in (essentially unlimited) long-term memory. This is why experts can hold far more "in mind" than novices: they chunk. [PhilPapers](#) [PubMed Central](#)

Learning vs performance (Bjork). As above: current performance and feelings of fluency are unreliable indices of long-term learning. Conditions that depress performance during practice (spacing, interleaving, retrieval, varied conditions) often *enhance* later retention and transfer — "desirable difficulties." Conversely, easy, massed, predictable practice inflates performance and confidence while undermining durable learning (Soderstrom & Bjork 2015, *Perspectives on Psychological Science*). [Percipio Global Ltd.](#)

Metacognition and accurate self-monitoring. Effective self-regulated learning depends on accurately judging what you do and don't know. The EEF's Teaching and Learning Toolkit rates metacognition and self-regulation among the highest-impact, lowest-cost approaches (about +7 months' additional progress on average), though this evidence base is largely from school-age learners and mixes correlational and intervention data. The key adult skill is calibrating your judgments of learning against objective feedback — which is exactly what retrieval practice provides. [Cloudfront + 2](#)

The illusion of fluency / fluency trap. Judgments of learning are strongly driven by processing fluency — the ease with which material is read or comes to mind — even when fluency does *not* predict actual recall (Koriat & Bjork; Carpenter et al. 2013, who showed a fluent, polished lecturer raised students' confidence in their learning without improving it). Re-reading and highlighting are dangerous precisely because they manufacture fluency. The antidote is to test yourself: failed or effortful retrieval gives you an accurate read on what you haven't yet learned.

[Structural Learning + 2](#)

Sleep and memory consolidation. Sleep actively consolidates newly learned material, and there is consistent evidence that sleeping after learning reduces forgetting. In one controlled study (Cousins, Teo, Tan, Wong & Chee 2021, *SLEEP* 44(3)), "sleep significantly enhanced consolidation of factual knowledge ($p = 0.01$, $d = 0.72$), but groups did not differ in their capacity

to relearn the materials ($d = 0.10$)"; Berres & Erdfelder's (2021, *Psychological Bulletin*) meta-analysis confirms an episodic-memory sleep benefit. A minority of researchers argue sleep's role is sometimes overstated (it may be neither strictly necessary nor sufficient). Still, the practical advice is robust: protect sleep, especially after heavy study, and avoid all-nighters that sacrifice consolidation. [Portland Press](#) [ScienceDirect](#)

The role of prior knowledge. Prior knowledge is one of the strongest determinants of new learning. It lowers intrinsic load (familiar elements are pre-chunked), enables richer elaboration, and is why the same instruction can be ideal for a novice and redundant for an expert (expertise reversal). Activating prior knowledge before studying new material — and filling gaps — is a high-value, evidence-based move (a central principle of the Deans for Impact *Science of Learning* report, whose second edition was released in May 2026).

5. Caveats on the evidence base

Most foundational lab studies used word lists, paired associates, or short texts with undergraduate participants over short retention intervals. The effects generally *do* replicate in authentic classrooms (Agarwal et al. 2021 for retrieval; classroom spacing and interleaving studies exist), but classroom effect sizes are typically smaller and more variable than lab effects, and long-term, cumulative-assessment effects are sometimes not significant (e.g. the year-long interleaving maths trial above). Evidence specific to *postgraduate adult* learners is thinner than for schoolchildren and undergraduates; the EEF months-of-progress figures in particular come from school populations and should not be read literally for Masters students. The underlying mechanisms (working memory limits, retrieval, spacing, consolidation) are general features of human cognition and almost certainly apply to adults, but exact magnitudes for your context are uncertain. Note too that some interleaving meta-analyses draw heavily on student samples whose homogeneity may inflate effect sizes.

Recommendations

Start here (highest return, do immediately):

1. **Replace re-reading and highlighting with retrieval practice.** After each lecture/reading, do a closed-book brain-dump, then check and correct. Build a flashcard deck for key terms, frameworks, and relationships.
2. **Space it out.** Plan term-long review from week one; revisit each topic several times at increasing intervals. Use a spaced-repetition app so scheduling is automatic. Treat cramming as a last resort, not a plan.
3. **Test under realistic conditions.** Do past papers and practice problems without notes and to time. Use the results — especially your failures — as an honest map of what you don't yet know (defeating the fluency trap).

Layer in next: 4. **Self-explain and elaborate** as you read: ask why/how, connect to prior knowledge, explain steps aloud. Especially valuable for the conceptual, integrative demands of Masters work. 5. **Interleave** problem types and confusable concepts in quantitative and category-heavy subjects. 6. **Use worked examples when new to a topic, then fade them** as you become fluent — match support to your expertise in that specific area. 7. **Build genuine dual-coded materials** (integrated diagrams/concept maps), not decorative ones.

Protect the foundations: 8. **Sleep** after studying; don't trade consolidation for extra low-quality hours. 9. **Activate prior knowledge** before tackling new material; deliberately fill prerequisite gaps. 10. **Calibrate your metacognition:** if a strategy feels easy and pleasant, be suspicious; if it feels effortful but productive, you're probably learning.

Benchmarks that should change your approach: If your performance on *delayed, closed-book* practice questions is rising, your strategy is working — keep going. If you feel confident but score poorly on unseen application/inference questions, you are in the fluency trap: shift from re-reading toward retrieval and self-explanation. If interleaving feels chaotic and your accuracy collapses, you may be mixing genuinely unrelated material or lack prerequisite knowledge — temporarily revert to worked examples and blocked practice until basics are secure, then re-interleave.

Caveats

- Effect sizes cited (Cohen's d / Hedges' g roughly 0.4–0.7 for the strong techniques) are averages with wide variation; your results depend on material, implementation, and how the strategy is used.
- The strongest evidence is for retrieval and spacing; interleaving, dual coding, and worked examples are real but conditional, and their benefits can vanish or reverse if misapplied.
- Lab-to-classroom generalisation is good but imperfect, and postgraduate-specific evidence is limited; treat the principles as well-founded and the precise numbers as indicative, not guaranteed.
- Where sources conflict (e.g. retrieval-practice effect size ~ 0.51 vs ~ 0.70 depending on comparison condition and format; whether sleep is "essential" for consolidation), I have reported the range rather than forcing a single figure.